

ArgoGuard AI

Intelligent Agricultural Co-Pilot & Resiliency Cockpit
Comprehensive Architecture & Design Specification

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Target System:	ArgoGuard Smart Core Infrastructure
Classification:	Internal Engineering Specification

1. Solution Overview

ArgoGuard AI is an advanced, enterprise-grade smart farming ecosystem and dual-frontend intelligent co-pilot. Engineered to bridge the operational gap between digital cloud analysis and physical field management, the platform coordinates multimodal computer vision, live meteorological feeds, automated IoT field hardware, and transactional state overrides. ArgoGuard AI safeguards crops, optimizes irrigation, and ensures agricultural operational resilience.

By integrating the power of high-confidence multi-agent reasoning with real-time feedback loops, the platform actively mitigates risk. Farm operators receive continuous visual and structural assurance that treatments are perfectly aligned with dynamic weather requirements and that physical hardware faults will never compromise field stability.

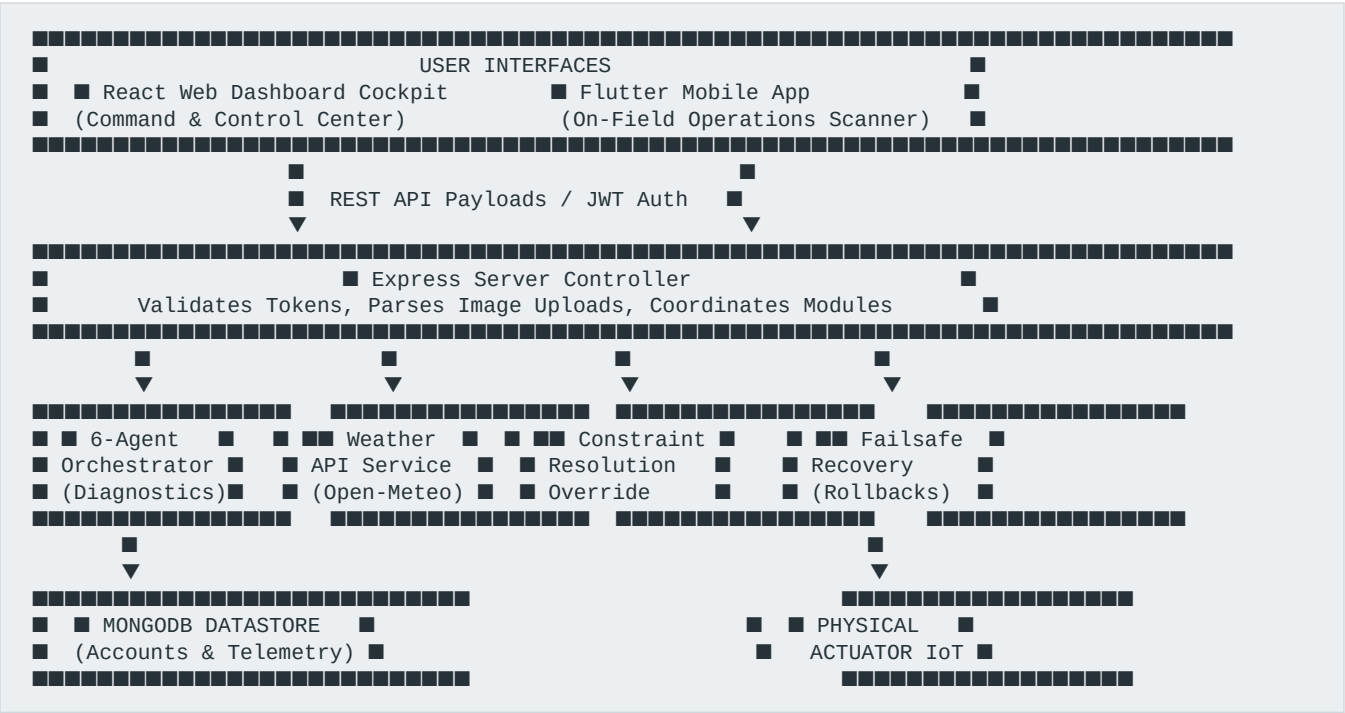
■ Core Value Proposition

- **Intelligent Diagnostics:** Leverages Google Gemini Multimodal APIs to analyze visual pathology with expert-level precision.
- **Contextual Safety:** Overrides irrigation and spray treatments dynamically if local weather signals pose contagion or run-off risks.
- **Transactional Failsafes:** Restores stable hardware baselines automatically in the event of controller connection drops, avoiding toxic over-spraying.

2. Decoupled Client-Server Architecture

The system architecture is structured as a robust, decoupled three-tier pattern. This isolates frontend presentation layers from the orchestrator logic and state management services, ensuring high horizontal scalability and offline reliability.

2.1 System Architecture Flow



2.2 Subsystem Details

A. React Web Dashboard Cockpit

Located inside the '/frontend' folder, this client represents the command cockpit for smart farm managers. Designed with React v19, Vite, and Tailwind CSS v4, it provides secure email credentials integrated with a 6-digit MFA OTP system, a 6-agent real-time diagnostics runner, visual before/after comparison sliders, and interactive hardware failure simulation panels.

B. Flutter Mobile App

Located in the workspace root, the Flutter mobile app is the physical field companion for farm workers. It leverages Provider state management reactive classes (AuthProvider, AnalysisProvider, WeatherProvider), supports direct camera pathology captures, uses geolocator to alert workers of high pesticide run-off weather risks, and features persistent local cache fallbacks.

3. The 6-Agent Diagnostic Orchestrator

The flagship service in the ArgoGuard AI backend is the custom Multi-Agent Orchestrator (multi_agent_orchestrator.js). To make sound decisions, the orchestrator divides critical tasks into a sequence of six specialized reasoning steps. This ensures transparent, auditable decision paths:

Step	Agent Name	Primary Domain Responsibility	Data I/O Mapping
1	Input Aggregator	Collects telemetry fields, matches crop image files, and polls local geolocation sensors.	Input: Image file, coordinates Output: Unified JSON payload
2	Disease Analyst	Queries Google Gemini Vision API to classify pathogen species, confidence rating, and suggested therapies.	Input: Image file Output: Pathogen ID, confidence, therapies
3	Risk Evaluator	Evaluates meteorological conditions (wind speed, rain) to assess disease spread dynamics.	Input: Local weather indices Output: Contagion percentage
4	Constraint Planner	Constraint Rules Audit: Audits suggestions against safety rules (e.g., overrides spray commands if high wind is reported).	Input: Suggested treatment, weather Output: Overridden, safe treatment plan
5	Action Executor	Constructs transactional payloads and issues dispatch signals to IoT smart valves or drone hubs.	Input: Approved treatments Output: Actuator codes & target statuses
6	Recovery Monitor	Listens for hardware feedback. If faults occur, manages retries, engages fallback suppliers, or initiates a state rollback.	Input: Hardware telemetry, latency Output: Safe hardware state lock & alerts

4. API Endpoint Specifications

The Node.js Express server exposes a secure, token-protected RESTful API. All endpoints require JWT Bearer headers and incorporate verification checks:

Method	Endpoint Path	Controller Reference	Auth	Resiliency Strategy
POST	/api/auth/register	auth_controller.js	None	Creates account, generates MFA token
POST	/api/auth/login	auth_controller.js	None	Returns JWT bearer token
POST	/api/auth/send-otp	auth_controller.js	None	Dispatches simulated email MFA code

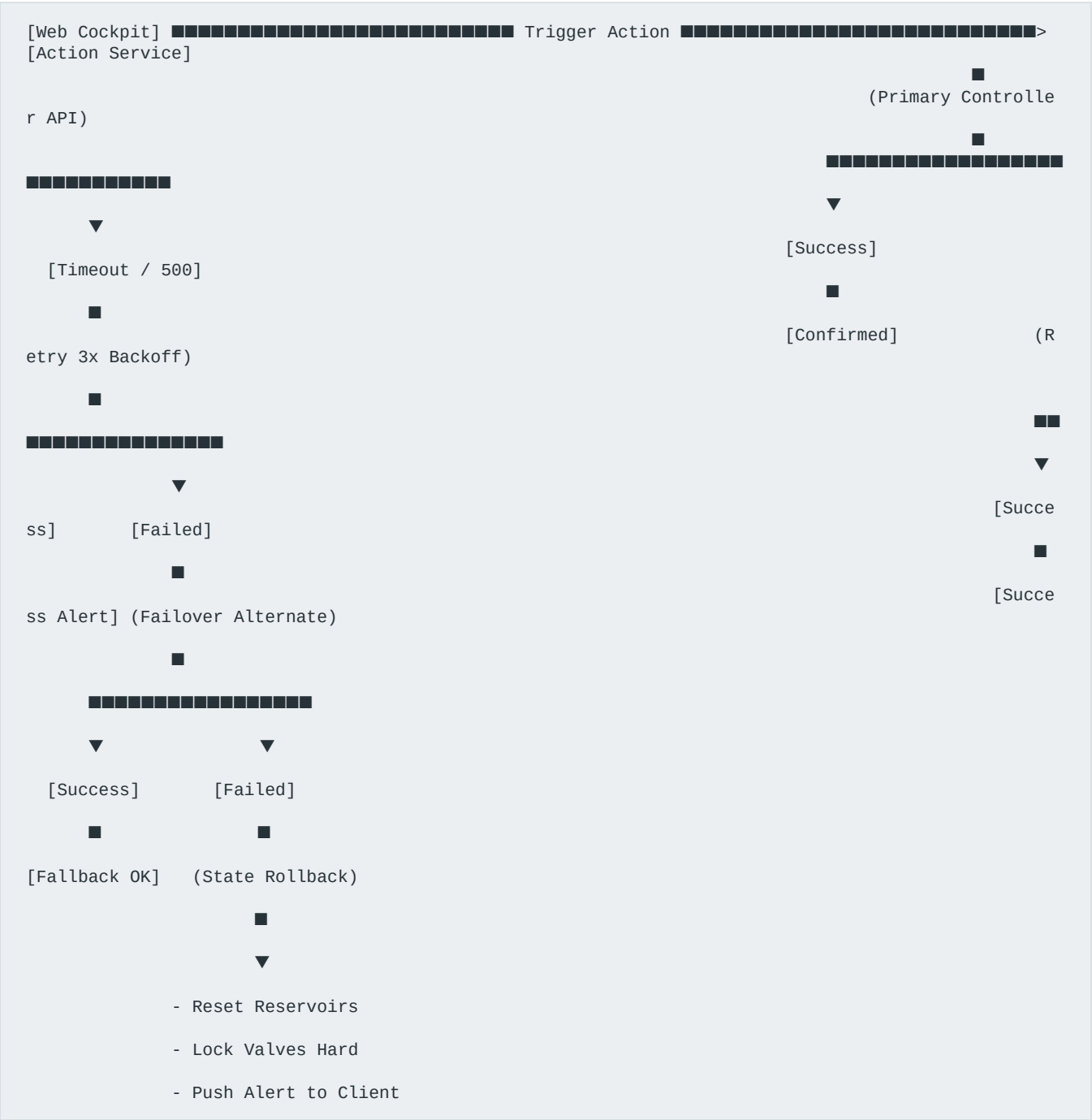
POST	/api/auth/verify-otp	auth_controller.js	None	Validates code; flags user verified
GET	/api/auth/me	auth_controller.js	JWT	Returns profile from authenticated JWT
POST	/api/upload	upload_controller.js	JWT	Handles multipart file streams via Multer
POST	/api/analyze	analyze_controller.js	JWT	Executes full 6-Agent Orchestrator Pipeline
GET	/api/weather	weather_controller.js	None	Connects to real-time Open-Meteo weather
POST	/api/actions	actions_controller.js	JWT	Dispatches hardware; tracks mock timeouts
GET	/api/logs	logs_controller.js	JWT	Streams step-by-step diagnostic telemetry traces
GET	/api/dashboard	dashboard_controller.js	JWT	Pulls charts history & flashing alerts data

5. Resiliency, Failsafe & Rollback Protocols

To prevent expensive crop loss or chemical leaks, ArgoGuard AI implements a robust, stateful Failsafe Recovery service (`recovery_service.js`). The system isolates faults and triggers active defenses in sequence:

- 1. Fault Detection:** The Action Service issues a request to a hardware valve or sprinkler controller.
- 2. Retries with Backoff:** If the connection experiences transient timeouts or 500 errors, the engine attempts up to 3 retries using an exponential backoff algorithm.
- 3. Redundant Controller Routing:** If retries are exhausted, the system attempts to route the action via an alternate, independent physical controller interface.
- 4. Stateful System Rollback:** If all controllers are unreachable, the Recovery Agent marks the event as an emergency [ROLLBACK] and resets reservoirs, sends safety overrides to close physical actuator valves, details the exact telemetry path in log tables, and dispatches high-priority push notifications.

5.1 Failsafe Sequence Flow



6. Development & Deployment Guidelines

ArgoGuard AI is designed for seamless, multi-environment deployments. Follow these commands to spin up a local instance for development, testing, or sandboxed simulations:

```
# 1. Start the Express Backend API Server
cd backend
npm install
npm run dev
# 2. Launch the React Web Cockpit
cd ../frontend
npm install
npm run dev
# 3. Run the Flutter Mobile Companion
cd ..
flutter pub get
flutter run
```

7. Architectural Summary

ArgoGuard AI provides a highly advanced, secure, and production-ready architecture. By modularizing crop pathology scanning and multi-agent reasoning, the system easily absorbs new sensor feeds, actuator drivers, and cloud intelligence models. Backed by strict constraint auditing, automatic retries, and transactional rollbacks, ArgoGuard AI sets a new standard of reliability for the future of digital farming.